

Traditional Methods of HIIT Programming

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Several approaches for prescribing high-intensity interval training (HIIT) have been developed, with the aim directed at ensuring athletes reach the required exercise intensity of their training sessions in a somewhat controlled and individualized way. While incremental laboratory-based test parameters may be the most accurate way to calibrate an HIIT session, a number of field-based maximal-critical speed and power analysis tools and methods may be more practical, objective, and effective for achieving the desired training and performance outcome. In this chapter, we describe the various traditional approaches used by coaches, scientists, and practitioners to calibrate interval training intensity. Procedures may include a rating of perceived exertion (RPE)-based prescription, the maximal aerobic speed and power-based method, the 30-15 intermittent fitness test, anaerobic speed reserve measures, all-out sprint training, the track-and-field or team sport approaches, and also heart rate and power meter-based approaches.

RPE-Based Prescription

Prescribing the intensity of an HIIT session using the RPE method (55) is highly attractive due to its simplicity. This method does not require the monitoring of heart rate, speed, or power. A coach can just instruct, “You’re running 4 × 4 min hard, with

2 min walk recovery intervals.” It’s a versatile method that can be used anywhere. Coaches have used this method more than any other for the last century to prescribe independent variables such as the duration or distance of work and relief intervals (105). Importantly, the athlete can self-regulate his exercise intensity based on how the exercise feels, the athlete’s brain being the best sensor globally of how in balance his body is (95). In the laboratory, the intensity selected is typically the maximal sustainable exercise intensity perceived as hard to very hard (i.e., ≥ 6 on a CR-10 Borg scale and ≥ 15 on a 6-20 scale) and is based on the athlete’s experience in the session, the session goal, and external considerations related to training periodization. While the specific roles (or contributions) played by varying biological afferents and other neurocognitive processes involved with the selection of an exercise intensity based on effort are debated (see viewpoint/counterpoint (81)), RPE responses may reflect “a conscious sensation of how hard, heavy, and strenuous the exercise is” (82), relative to the combined physiological (44), biomechanical, and psychological (83) stress or fatigue imposed on the body during exercise (110).

In practice, the first benefit of RPE-guided HIIT sessions (105, 106) is that they do not require any knowledge of the athletes’ fitness level, making them highly convenient to use. Test results are not

required. Therefore the RPE method is considered a universal exercise regulator, irrespective of the exercise mode and variations in terrain and environmental conditions. Interestingly, the RPE-based prescription method has been shown to promote the same physiological outcomes as a heart rate-calibrated program over 6 wk in young women (41).

However, the method does not allow for the precise manipulation of the physiological response to a given HIIT session, which could limit the ability to target a specific adaptation. For example, Seiler and Sylta (107) compared the physiological and RPE responses in 63 well-trained cyclists performing three different maximal-effort HIIT sessions differing only in work bout and accumulated duration. Sessions were prescribed as 4×16 min, 4×8 min, or 4×4 min, all separated by 2 min recovery periods. Sessions (mean(SD)) were performed at 95(5), 106(5), and 117(6)% of 40 min time trial power during 4×16 , 4×8 , and 4×4 min sessions, respectively. Percent peak heart rate of the efforts were 89(2)%, 91(2)%, and 94(2)%, with blood lactate concentration at 4.7(1.6), 9.2(2.4), and 12.7(2.7) mM, respectively. While all sessions were instructed to be maximal efforts, RPE increased with lowered volumes of accumulated work duration and with higher intensity. This highlights a limitation of the RPE method, with the duration of the prescribed effort influencing the physiological outcome or effect. While the RPE method is popular and often used with objective monitoring, use of the RPE method solely without other markers of performance (i.e., speed, power) and its related physiological response (i.e., HR, lactate) can present a level of uncertainty within a coach's and practitioner's programming puzzle.

Maximal Aerobic Speed and Power

The speed and/or power associated with the body's maximal oxygen uptake ($\dot{V}O_{2\max}$; so-called $v/p\dot{V}O_{2\max}$ or maximal aerobic speed/power [MAS/MAP] (8, 69)) has been shown to be a useful reference intensity for programming HIIT (6, 7, 76). To be clear, $\dot{V}O_{2\max}$ is described as the greatest amount of oxygen that can be used at the cellular level for the entire body, and is related primarily to the ability of the heart and circulatory system to transport oxygen and the ability of body tissues to use it

(expanded on in chapter 3). It makes intuitive sense to be mindful of the performance level associated with this factor, as $\dot{V}O_{2\max}$ has been consistently correlated with distance running (94) and cycling performance (63). The attractiveness of the $v/p\dot{V}O_{2\max}$ method is that it represents an integrated measure of both $\dot{V}O_{2\max}$ and the energetic cost of running/cycling into a single factor, and might represent an athlete's peak locomotor ability (figures 2.1 and 2.2) (8). Since $v/p\dot{V}O_{2\max}$ is theoretically the lowest speed/power needed to elicit $\dot{V}O_{2\max}$, it makes intuitive sense for this marker to represent an ideal reference for training (8, 76, 90).

The $v/p\dot{V}O_{2\max}$ can be determined, or estimated, using a number of different ways. Some of these include the following:

1. For running, the linear relationship between $\dot{V}O_2$ and running speed established at a submaximal speed (46) or the individual cost of running is used to calculate a theoretical running speed for a given $\dot{V}O_{2\max}$, either with (53) or without (74) resting $\dot{V}O_2$ values.
2. Direct measurement (i.e., pulmonary gas exchange (9)) during ramp-like incremental running or cycling tests to exhaustion is used, either on a track or treadmill or by using an ergometer (78). On the track, the University of Montreal track test (UM-TT (79)) is the protocol most commonly used with team sport athletes (16, 56), although the Vam-Eval has also been used (40) and differs from the UM-TT in that it has smoother speed increments and shorter inter-cone distances. The Vam-Eval has received growing interest since it may be easier to administer in young populations and non-distance running specialists (25, 89). For cyclists, progressive tests to exhaustion on a cycle ergometer using fast ramp step increases (i.e., 30 W/min) that elicit $\dot{V}O_{2\max}$ and its associated power output (78) can be used. Similar methods can be used for evaluation with other modes of exercise, including rowing (43) and cross-country skiing (114).

Since the true $v/p\dot{V}O_{2\max}$ during incremental tests requires $\dot{V}O_2$ measures to determine the lowest speed/power that elicits $\dot{V}O_{2\max}$ (generally defined as a plateau in $\dot{V}O_2$ or an increase less than $2.1 \text{ mL} \cdot \text{min}^{-1} \cdot \text{kg}^{-1}$ despite an increase in running speed of 1 km/h (9) or power output increase of 30 W (78)), the final (peak) incremental test speed/power reached at the end of these tests (V/P_{IncTest})

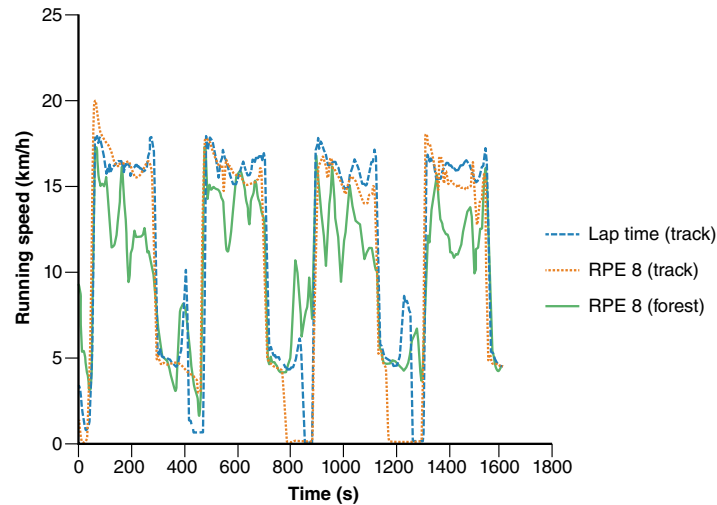


Figure 2.1 Running speed during long-interval HIIT with pace dictated through audio feedback (blue line), where beeps sound at appropriate time intervals to guide pace in a team sport athlete running at 95% of $V_{IncTest}$ or RPE on either a track (orange line) or in a forest (green line). Note that running at an RPE of 8 (10 pt scale) results in very similar speed patterns compared to using the more controlled audio feedback on the track. Interestingly, running speed shows large variations in relation to changes in running terrain in the forest, likely to be independent of metabolic strain (RPE 8). Thus, RPE can allow athletes to regulate running speed at the required level of exertion relative to more controlled means.

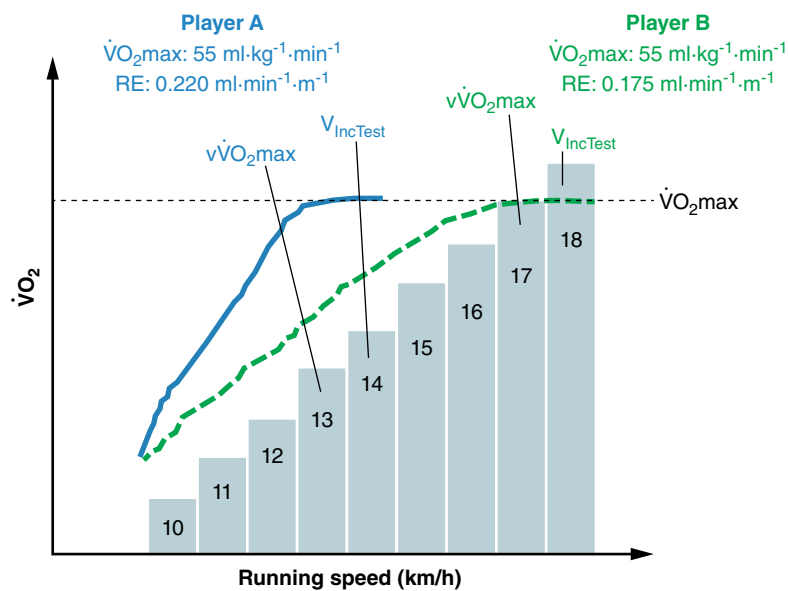


Figure 2.2 Time course of oxygen uptake ($\dot{V}O_2$) during an incremental test for two players with similar $\dot{V}O_{2max}$ but with different running economy. Player A (blue) presents a poor economy, while player B (green) a very good running economy. The energetic cost of running being greater for player A, the velocity at which he reaches $\dot{V}O_{2max}$ ($v\dot{V}O_{2max}$) and therefore the final velocity ($V_{IncTest}$) of the test are both substantially lower than for player B.

Adapted from M. Buchheit, "The 30-15 Intermittent Fitness Test: 10 Year Review," *Myorobie Journal* 1 (2010): 1-19.

is only an approximation of $v/p\dot{V}O_{2\max}$. These two distinct speeds/powers are strongly correlated ($r > 0.90$ (79)), but V/P_{IncTest} can be 5% to 10% greater than $v/p\dot{V}O_{2\max}$, with individuals possessing greater anaerobic reserves presenting generally a greater $v/p\dot{V}O_{2\max} - V/P_{\text{IncTest}}$ difference (discussed subsequently).

Despite the subtleties, a number of indirect methods can be used to predict $v/p\dot{V}O_{2\max}$ that may be more practical for field testing, including:

1. Running: A 5 min exhaustive run (5), since the average time to exhaustion at $v\dot{V}O_{2\max}$ has been reported to range from 4 to 8 min (8, 70). The $v\dot{V}O_{2\max}$ calculated from this test has been shown to be largely correlated with the V_{IncTest} reached in the UM-TT ($r = 0.94$) and on a ramp treadmill test ($r = 0.97$) (5), while being slightly (i.e., 1 km/h range) slower and faster than these velocities, respectively. However, the $v\dot{V}O_{2\max}$ estimated via the 5 min test may be influenced by pacing strategies and likely is most applicable to use with trained runners able to run at $v\dot{V}O_{2\max}$ for approximately 5 min. However, this 5 min test is used more in the research setting, and a comparable test in the field, as used with Australian rules football teams, might involve 2 km time trials (typically about 6 to 7 min) that are far easier to administer (see chapter 25).

2. Cycling: A 4 min all-out cycling power test in either the laboratory or in the field using a power meter, since the power output sustainable over 4 min, or the maximal mean power output (MMP), tends to be within close range of the $p\dot{V}O_{2\max}$ (99) (75, 77). Practically speaking for cycling coaches and practitioners, the simple process of monitoring cycling performance with a power meter and related software programs enables the detection of $p\dot{V}O_{2\max}$ or V/P_{IncTest} , simply called 4 min MMP, and is assessed either in the field or specifically using a power profile test (99) (see chapter 15).

3. Rowing: A 2 km rowing time trial (about 6 to 7 min) on a Concept II ergometer, which is also performance specific as most rowing events are determined over this competition distance (39) (see chapter 16).

Technically speaking, it's important to remember that V/P_{IncTest} or $v/p\dot{V}O_{2\max}$ can be method (69) and protocol dependent (91). An estimated $v\dot{V}O_{2\max}$ (47, 53) has been shown to be lower than a measured

$v\dot{V}O_{2\max}$ (9) and V/P_{IncTest} (8, 62). Additionally, irrespective of the method used to determine $v/p\dot{V}O_{2\max}$, protocols with longer stage durations tend to elicit lower speed/power values (91), while larger speed/power increments (resulting in shorter tests) may result in higher speed/power values, the anaerobic capacity of the individual being the confounding variable in the assessment. Similarly, $v/p\dot{V}O_{2\max}$ can be inversely related to the terrain or treadmill slope (98). Endurance-trained athletes are likely able to tolerate longer stages and, therefore, less likely to present impairments in $v/p\dot{V}O_{2\max}$ with variations in protocol (3).

Nevertheless, Manoel et al. (80) compared the effects of 4 wk of training prescribed using the time limit associated with either V_{IncTest} or $v\dot{V}O_{2\max}$ in moderately trained endurance runners. The authors showed similar improvements in both groups, suggesting that use of V_{IncTest} can be recommended in the field, as it can be easily self-administered on a treadmill. V_{IncTest} can be determined in the field with the UM-TT (79) or the Vam-Eval (40), since, in addition to not requiring sophisticated apparatus (i.e., metabolic carts), the tests conform with the incremental rate of 0.5 km/h per minute recommended for accurate $v\dot{V}O_{2\max}$ determination (4) and account for the anaerobic contribution necessary to elicit $\dot{V}O_{2\max}$.

Global positioning system (GPS) technology, though not without its limitations, is becoming more reliable and is certainly the most practical method for remote monitoring. As with cycling power meters, GPS systems can be integrated with online software platforms to enable seamless integration and calculation of V_{IncTest} in the field. For cycling training prescription without access to sophisticated equipment such as metabolic carts, $p\dot{V}O_{2\max}$ and P_{IncTest} may be most practically estimated in the field using a calibrated power meter to enable determination of 4 min MMP (99).

The reliability of $v/p\dot{V}O_{2\max}$ and V/P_{IncTest} (as examined using coefficients of variation, CV) has been shown to be good: 3% for $v\dot{V}O_{2\max}$ in moderately trained middle- and long-distance runners (92); 3.5% for UM-TT V_{IncTest} in moderately trained athletes (79); 3.5% (90% confidence limits: 3.0;4.1 (Buchheit, unpublished results)) for Vam-Eval V_{IncTest} in 65 highly trained young football players; 2.5% and 3% for treadmill V_{IncTest} in well-trained male distance (104) and recreational runners (62), respectively; 1%-2% for the 5 min test in a heterogeneous

sporting population (45); and about 2% in well-trained cyclists (77).

However, it is worth noting that using $v/p\dot{V}O_{2\max}$ or V/P_{IncTest} as the reference running speed or power output is essentially suitable for long (2 to 6 min) intervals performed around $v/p\dot{V}O_{2\max}$ (90% to 105%). For sub- and supramaximal training intensities, however, the importance of other physiological attributes should be considered. For instance, endurance capacity (the so-called anaerobic threshold, or capacity to sustain a given percentage of $v/p\dot{V}O_{2\max}$ over time (11)) and anaerobic power/capacity (10) are likely to influence time to exhaustion, and in turn, the physiological responses.

Heart Rate-Based Prescription

Today the monitoring of an athlete's heart rate is probably the most commonly measured physiological marker used to control or measure exercise intensity in the field (19). Setting an exercise intensity using HR zones is well suited to prolonged and submaximal exercise bouts; however, its effectiveness for controlling or adjusting the intensity of an

HIIT session may be limited. Indeed, HR alone cannot inform on the intensity of physical work performed above $v/p\dot{V}O_{2\max}$, which represents a large proportion of HIIT prescriptions (6, 7, 76). Additionally, while HR is expected to reach maximal values (>90%-95% $HR_{\max}$) for exercise at or below $v/p\dot{V}O_{2\max}$, this is not always the case, especially for very short (<30 s) (92) and medium-long (i.e., 1-2 min) (106) intervals. This is related to the well-known HR lag at exercise onset, where HR is much slower to respond compared with the $\dot{V}O_2$ response (figure 2.3) (42). Further, HR inertia at exercise cessation (i.e., HR recovery) can also be problematic for prescribing in this context, since this can create an overestimation of the actual work or physiological load that occurs during recovery periods (106). It has been shown that substantially different exercise sessions, as assessed by accumulated blood lactate levels during run-based HIIT (105) or by running speed during matches (SSG) (86), can have relatively similar overall average HR responses (figure 2.4). Thus, the temporal dissociation between HR, $\dot{V}O_2$, blood lactate levels, and work output during HIIT limits our ability to accurately

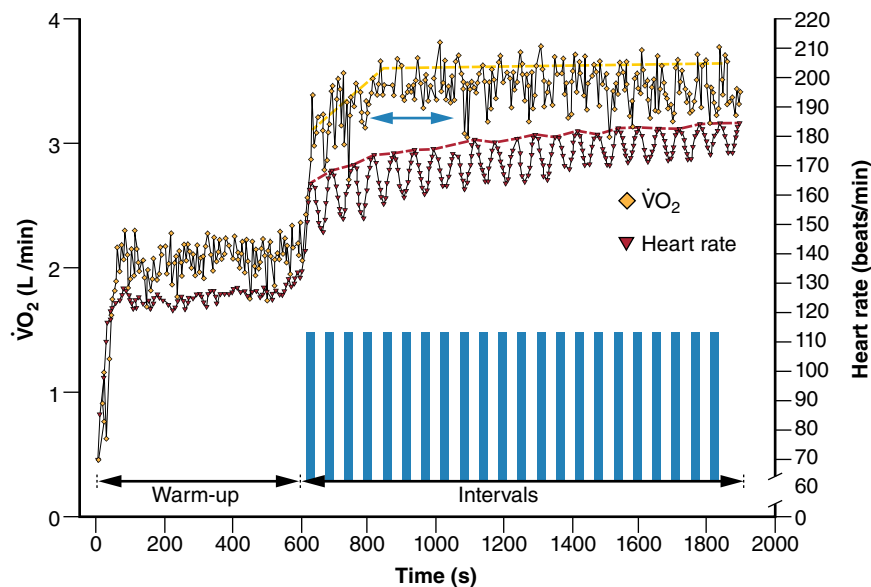


Figure 2.3 Oxygen uptake ($\dot{V}O_2$) and heart rate (HR) responses during HIIT with short intervals (30 s at 105% $v\dot{V}O_{2\max}$ interspersed with 30 s at 60%). Note the dissociation between $\dot{V}O_2$ and HR responses at the start of the exercise, i.e., HR adapts much slower than $\dot{V}O_2$, revealing its poor value to inform directly on metabolic demands.

Adapted by permission from A.W. Midgley, L.R. McNaughton, and S. Carroll, "Reproducibility of Time at or Near $\dot{V}O_{2\max}$ during Intermittent Treadmill Running," *International Journal of Sports Medicine* 28 no. 1 (2007): 40-47. © Georg Thieme Verlag KG

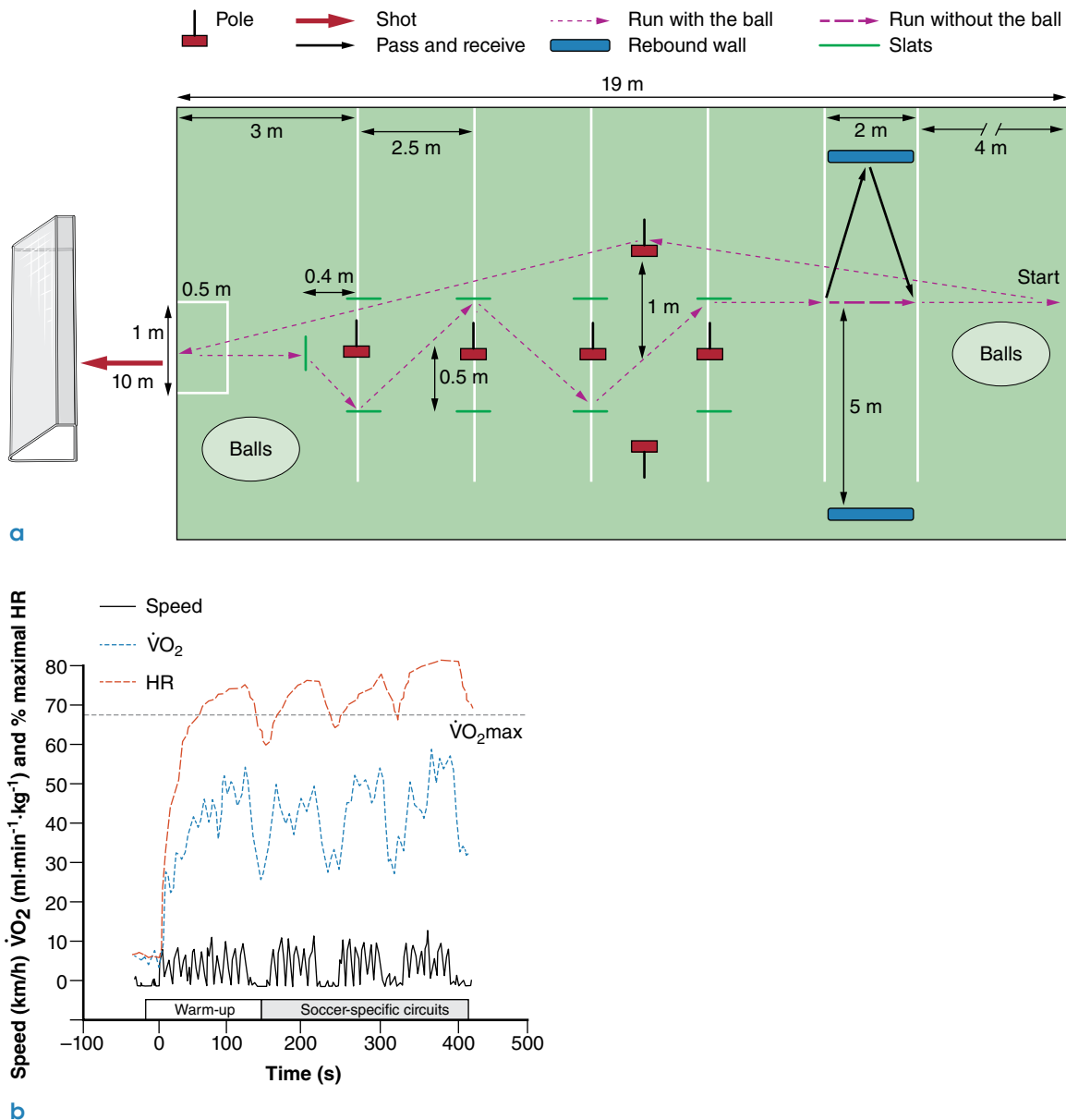


Figure 2.4 (a) Running circuit in which players repeat actions with the ball. (b) Oxygen uptake ($\dot{V}O_2$), heart rate (HR), and speed during the warm-up and the three exercise bouts in a representative player.

Adapted by permission from M. Buchheit, C. Manouvrier, J. Cassirame, and J.B. Morin, "Monitoring Locomotor Load in Soccer: Is Metabolic Power, Powerful?" *International Journal of Sports Medicine* 36 no. 14 (2015): 1149-1155.

estimate intensity during HIIT sessions using HR alone. Instead it should be used as one of a number of markers in assessment (see chapters 8 and 9). Further, it is difficult to imagine how an athlete would practically control or adjust exercise intensity *during* an interval, especially for athletes running at high speed, where viewing HR from a watch is difficult. To investigate this directly, Rabinini and Buchheit (100) compared HIIT prescription over 4 to 5 wk in two groups of young soccer

players, one using HR-based prescription (HIIT using 90%-95% HR_{max}) and the other using the speed reached at the end of the 30-15 intermittent fitness test (V_{IFT}) as the reference for running intensity (expanded on shortly). All tactical and technical training remained the same in both groups, and all HIIT sessions were completed twice per week, involving 3 sets of 3:30 duration. Results showed that using the V_{IFT} as a reference speed for HIIT programming produced greater high-intensity

intermittent running performance improvements compared with HR-based methods. Thus, HR monitoring over prescription per se is what occurs in practical settings in most sports. For example, in endurance sports, a session might be prescribed as a level 5 (L5) workout (5-zone intensity model), and in either case, the athlete will know from experience to perform these sessions at the appropriate pace or power output (RPE method). In this case, HR is simply monitored to observe the response. Similarly in team sports, it is common to use HR responses to standardized HIIT (with individual target running distances prescribed) as monitoring points throughout the season (i.e., fitness testing without formal testing, per se) (30). For example, Buchheit et al. (30) used this strategy with an Australian rules football team during a 4 min 15"15" HIIT session, revealing that fitness and the HR response to HIIT was maintained during their 2 wk Christmas break.

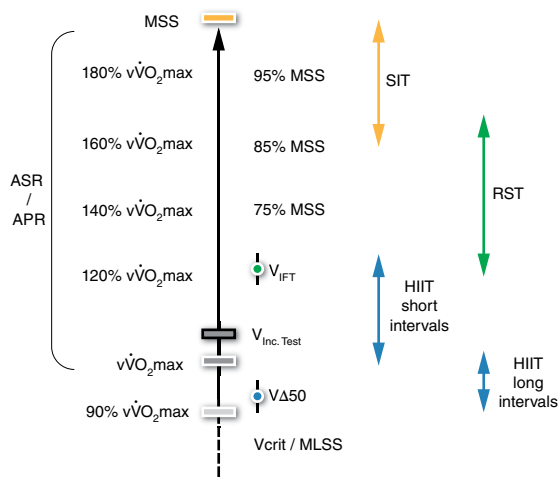


Figure 2.5 Intensity range used for the various HIIT formats. ASR: anaerobic speed reserve; APR: anaerobic power reserve; MLSS: maximal lactate steady state; MSS: maximal sprinting speed; RST: repeated-sprint training; SIT: sprint interval training; $\dot{V}O_{2\max}$: maximal oxygen uptake; $\dot{V}O_{2\max}$: minimal running speed required to elicit $\dot{V}O_{2\max}$; $V_{\Delta 50}$: speed halfway between $\dot{V}O_{2\max}$ and MLSS; V_{crit} : critical velocity; V_{IFT} : peak speed reached at the end of the 30-15 intermittent fitness test; V_{IncTest} : peak incremental test speed.

Adapted by permission of Springer Nature from M. Buchheit and P.B. Laursen, "High-Intensity Interval Training, Solutions to the Programming Puzzle: Part I: Cardiopulmonary Emphasis," *Sports Medicine* 43 no. 5 (2013): 313-338.

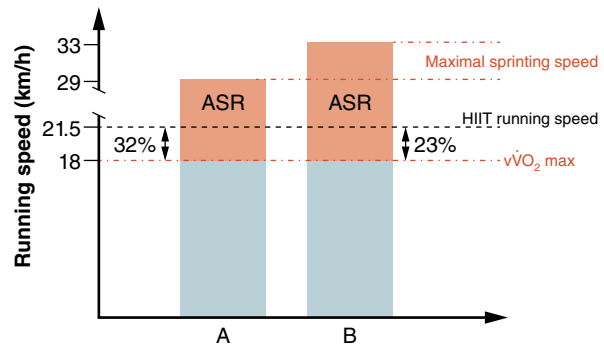


Figure 2.6 The anaerobic speed reserve (ASR) of two players having a similar $\dot{V}O_{2\max}$ but different maximal sprinting speed. During an HIIT session, player B with the greater ASR will work at a lower percentage of his ASR and will therefore present a lower exercise load compared to player A.

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Thus far we have been largely discussing HIIT programming methods that more or less apply to exercise performed above critical power but below $v/p\dot{V}O_{2\max}$. The following sections highlight the different options available for the prescription of supramaximal training (i.e., training at intensities greater than $v/p\dot{V}O_{2\max}$).

Anaerobic Speed Reserve

Anaerobic speed reserve (ASR) describes the exercise capacity between one's $v/p\dot{V}O_{2\max}$ and his maximal sprinting speed (MSS) or anaerobic peak power (i.e., MMP_{5s} ; figure 2.5). Consideration for an individual's ASR is often not fully taken into account by coaches and scientists in their training prescription. While track-and-field (running) coaches have indirectly used this concept for years to set the work interval intensity (see chapter 13), its scientific basis and interest was brought forth only two decades ago, when Blondel and coworkers (10) showed that time to exhaustion at intensities above $\dot{V}O_{2\max}$ were better related to the ASR and MSS than to $\dot{V}O_{2\max}$. Subsequently, research (35, 111, 112) showed using an empirical prediction model that the proportion of ASR used could determine performance during all-out efforts lasting between a few seconds and several minutes. While these studies have used continuous exercise, ASR

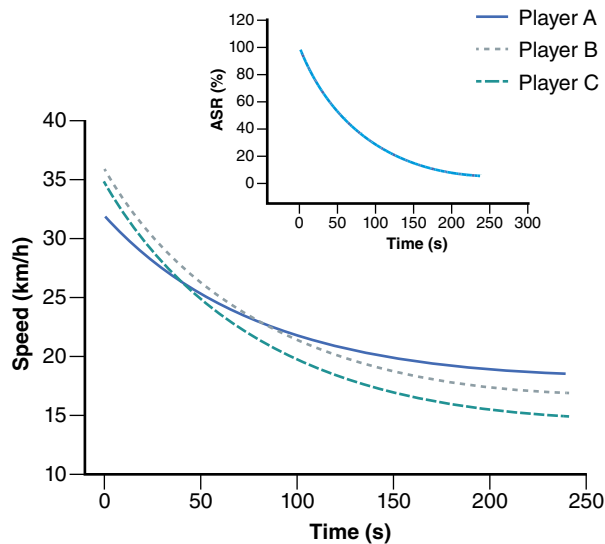


Figure 2.7 Predicted speed/duration relationship in three soccer players with different maximal sprinting (MSS) and aerobic (MAS) speeds and peak speed reached in the 30-15 intermittent fitness test (V_{IFT}). Player A: 32, 18, and 21 km/h for MSS, MAS, and V_{IFT} , respectively; player B: 36, 16, and 20.5 km/h; and player C: 35, 14, and 18.5 km/h.

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has only recently been considered in relation to high-intensity intermittent running (20, 24) and repeated-sprint performance (23, 26, 85, 109). In practice, two athletes can present with clearly different MSS ability, despite a similar $\dot{V}O_{2\max}$ (figure 2.6 (17)). If during an HIIT session they exercise at a similar percentage of $\dot{V}O_{2\max}$, as is generally implemented in the field (e.g., Dupont et al. (57)), the exercise will actually involve a different proportion of their ASR (figure 2.7), which results in a different physiological demand, and in turn, a different exercise tolerance (17). Therefore, it appears that, in addition to $\dot{V}O_{2\max}$, the measurement of MSS and ASR should be considered for individualizing training intensity during supramaximal HIIT (i.e., figure 2.6) (10, 17).

A limited number of methods for determining MSS and peak power are known, but these methods are unlikely to produce differences as great as those seen with $v\dot{p}\dot{V}O_{2\max}$ or $V/P_{IncTest}$ measurement (21). For MSS determination, the use of a flying 10 (27) or 20 m (12, 96) time is probably the most common

method of determination, although peak instantaneous speed can also be measured now with radar gun technology (73) or GPS (2). The difference between these methods is generally less than 2%, which is substantially lower than the possible 20% difference that can be observed for $\dot{V}O_{2\max}$ or $V_{IncTest}$ determination (21). Moreover, the typical error of measurement for MSS (1%) is clearly lower than that of $V_{IncTest}$ (3.5%) (22). Taken together, this suggests that determination of ASR is less likely to be affected by variations in MSS compared to $\dot{V}O_{2\max}$ and $V_{IncTest}$ estimation (21).

For peak cycling power, all-out cycling power output over 6 s can be reliably assessed in the laboratory (84), and in practice also can be carefully monitored in the field in cyclists using power meters (99). Doing so enables estimate of the anaerobic work capacity, or W' . The concept of W' is expanded on in chapter 4.

30-15 Intermittent Fitness Test (30-15 IFT) (<https://30-15ift.com>)

While using the ASR to individualize exercise intensity for supramaximal runs might represent an improvement over that of $\dot{V}O_{2\max}$ or $V_{IncTest}$, it still does not capture the overall picture of the different physiological variables of importance during team- or racket-based specific HIIT sessions or middle-distance running (103). In many sports, HIIT is performed indoors and includes repeated very short work intervals (<45 s). This implies that, in addition to the proportion of the ASR used, the responses to these forms of HIIT appear related to an individual's metabolic inertia (e.g., $\dot{V}O_{2\max}$ kinetics) at the onset of each short interval, physiological recovery capacities during each rest interval, and change of direction ability since indoor HIIT is often performed using shuttles (16, 17). Programming HIIT without taking these variables into consideration may result in sessions with different aerobic and anaerobic/neuromuscular demands, which prevents the standardization of training load, and may limit the ability to target specific physiological/neuromuscular adaptations (16). To overcome the limitations inherent with the measurement of $\dot{V}O_{2\max}$ and ASR, the 30-15 intermittent fitness test (30-15 IFT; figure 2.8) was developed for intermittent exercise and change of direction (COD)-based HIIT prescription (13, 16, 17).

THE 30-15 INTERMITTENT FITNESS TEST

Martin Buchheit

Everything started during the summer of 1999. As a strength and conditioning coach—and player—in a French handball team in Strasbourg (performing at the 4th French level), I was at this time using the University of Montreal track test (29) to evaluate the cardiovascular fitness of our players, and using the final velocity attained in this test (V_{UMTT}) to individualize running distances during interval training sessions (3, 24). For long intervals, this worked fine. Each player could cover a known distance based on their own capacity. This saw us performing well at 85%-90% of V_{UMTT} for 10–12 or even 15 min. However, later during the preparatory phase, when we started to work at higher intensities indoors (i.e., shuttle runs at intensity above V_{UMTT}), things became more complex. Some players were struggling, while for others the work was easy. That surprised me, since I was still individualizing running distance based on a player's physiological capacity (i.e., 120% of V_{UMTT}).

Across the team, players had different athletic and anthropometric profiles: some were tall, some were short, some were quick, and some were slow. This made me realize that the players' responses to HIIT with changes of direction (COD) were clearly related to many factors other than those evaluated with the UMTT track test (e.g., COD ability, inter-effort recovery abilities, anaerobic capacity, and speed). A new field test was clearly needed—one that could assess, in addition to maximal cardiorespiratory fitness, these additional factors. The idea of an intermittent incremental field test that included COD, taxing as well the anaerobic glycolytic system, and allowing higher maximal running speeds than the usual protocols, was conceived.

It took me more than a year of reading, attending talks, and experimenting with trials before the actual 30-15 IFT protocol was finished (7). Across different surfaces and in different venues, I have personally trialed more than 30 different versions of the test, based on different increments, stage durations, recovery periods, etc. Eventually, the 30-15 IFT was “born” in July 2000, using the following formula:

Repeated 40 m shuttle runs over 30 s, interspersed with 15 s of active (walking) recovery, with speed increments of 0.5 km/h per stage.

Today, the test is used all over the world in almost every possible sport. Due to the daily requests received from users running the test in their setting, I produced an app that anyone can use in their own setting. The 30-15 IFT App was released in August 2018 and received more than 2000 downloads in its first week. This continued high interest in the test from practitioners all around the world is the most rewarding gift I could have ever dreamed of 18 years ago.

The 30-15 IFT was designed to elicit maximal HR and $\dot{V}O_2$, but additionally provide measures of ASR, inter-effort recovery capacity, acceleration, deceleration, and COD abilities (15, 16, 32). The final speed reached during the test, V_{IFT} , is therefore a product of those abilities. In other words, the 30-15 IFT is highly specific, not to a sport, but to

the training sessions commonly performed in intermittent sports (17). While the peak speeds reached in the different Yo-Yo tests (1) (e.g., vYo-YoIR1 for the Yo-Yo intermittent recovery level) (1) have similar physiological requirements, only the V_{IFT} can be used accurately for training prescription. For instance, vYo-YoIR1 cannot be directly used for

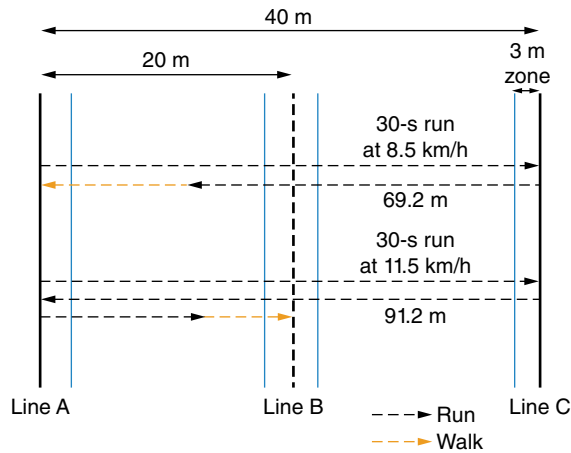


Figure 2.8 Area prepared for the 30-15 IFT and example of two intermittent runs. For the run at 8.5 km/h (about 69.2 m in 30 s), subjects start at line A, run to line C crossing line B and then return. After crossing line B again, they stop after 8.5 m and walk to line A during the 15 s recovery to be ready for the next stage. For the run at 11.5 km/h (about 91.2 m in 30 s), subjects start at line A, make one complete round trip, and stop after 9.5 m when going toward line B, then walk to line B during the 15 s of recovery for the next start. Note that calculation of targeted distances takes into account the time needed for the direction changes (for more, see <https://30-15ift.com>).

Reprinted from M. Buchheit, "The 30-15 Intermittent Fitness Test: 10 Year Review," *Myorobie Journal* 1 (2010): 1-19.

training prescription since, in contrast to V_{IFT} (13), its relationship with $V_{IncTest}$ (and $\dot{V}O_2$ max) is speed dependent (58). When running at $vYo-YoIR1$, slow and less-fit athletes use a greater proportion of their ASR, while fitter athletes run below their $\dot{V}O_2$ max (figure 2.9). Finally, V_{IFT} has been shown to be more accurate than $V_{IncTest}$ for individualizing HIIT with COD in well-trained team sport players (16), and its reliability is good, with the typical error of measurement (expressed as CV) shown to be 1.6% (95% CL, 1.4 to 1.8) (14). Also of note, since V_{IFT} is 2 to 5 km/h (15%-25%) faster than $\dot{V}O_2$ max or $V_{IncTest}$ (13, 17, 32), it is necessary to adjust the percentage of V_{IFT} used when programming. While HIIT is generally performed around $\dot{V}O_2$ max (i.e., 100%-120% (6, 7)), V_{IFT} constitutes the upper limit for these exercises (i.e., 100% except for very short intervals and all-out repeated-sprint training). Thus, the 30-15 IFT permits improved precision of programming by individualizing the intensity and duration of the prescribed interval bouts around

intensities ranging from 85% to 105% of V_{IFT} (28, 31, 33, 51, 93).

To offer a real-life example, if we had a player with a V_{IFT} of 19 km/h performing a 15 s/15 s HIIT run at 95% of V_{IFT} (rest interval: passive), his target distance would be $(19/3.6) \times 0.95 \times 15 = 75$ m (19 is divided by 3.6 to convert km/h to m/s, for convenience) (18). When it comes to implementing HIIT in the field using targeted shuttle run distances, in addition to the fraction of V_{IFT} used, additional time needed to implement changes of direction must also be considered to equalize to the straight-line run cardiorespiratory load equivalent. Indeed, adding CODs over the same shuttle distances and durations have been shown to substantially increase relative exercise intensity (52) (more related to the number of CODs and its effect on running speed) (29). Thus, the COD correction factor can vary between 3% and 30%. While robust scientific evidence may be lacking, in practice a player's height and training volume can be considered for individual adjustments, with smaller and more-trained athletes typically presenting better COD ability (29), thereby requiring a lower correction factor. Presently the correction factor lacks direct individualization and is based more on a player's average profile. As the adjusted difference is typically not greater than 1 to 2 m, the shuttle distance can be modified a posteriori, as needed, so splitting hairs worrying about precision here is not required. Figure 2.10 shows normal values for the 30-15 IFT test across various sports and playing standards (for more, see <https://30-15ift.com>).

All-Out Sprint Training

Repeated all-out sprinting efforts or repeated sprint training (RST; figure 2.11) are exactly that—all-out sprints performed over a given distance or duration (61, 72). In practical terms, such sessions can be divided into either short (3-10 s; RST) or long (30-45 s; sprints) duration sprints, the latter termed sprint interval training (SIT). Since such exercise is consistently performed all-out, it can be prescribed without the need to pretest the individual (i.e., $v/p\dot{V}O_2$ max is not specifically needed to calibrate the intensity).

We described the cardiorespiratory and muscle oxygenation responses to all-out sprint interval training (26; figure 2.11). In this study, well-trained cyclists performed 6×30 s all-out cycling sprints, each separated by 2 min of passive recovery. Despite a 21% decline in power output, we showed a progressive increase in muscle O_2 extraction with each suc-

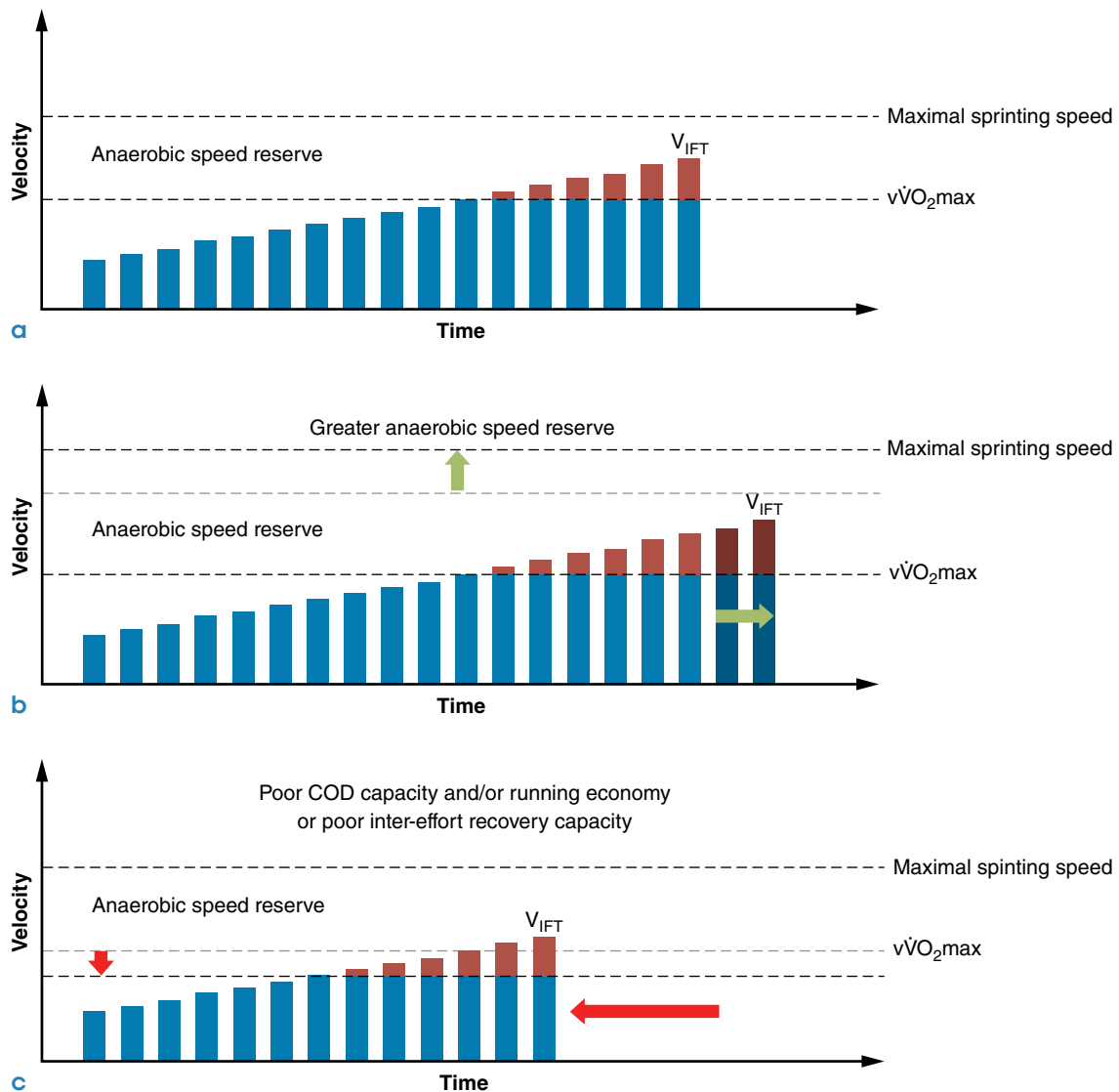


Figure 2.9 The importance of the anaerobic speed reserve (ASR) contribution toward reaching the final speed during the 30-15 intermittent fitness test (V_{IFT}). (a) Once the minimum speed required to elicit maximal oxygen uptake ($\dot{V}O_{2max}$) is reached, the additional energy provided during the remaining stages is derived mostly from anaerobic sources. Therefore, for a given $\dot{V}O_{2max}$, the greater maximal sprinting speed (MSS), and hence the ASR (or at least the greater the proportion used), the greater the number of supra- $\dot{V}O_{2max}$ stages that will be completed and the faster the V_{IFT} . (b) This player has a greater ASR and is able to reach, for a similar $\dot{V}O_{2max}$, two further stages during the 30-15 IFT (1 km/h). Since the ASR (or the proportion used) also influences what can be achieved during high-intensity intermittent runs, the use of the V_{IFT} , and not $\dot{V}O_{2max}$, enables the programming of similar (anaerobic and neuromuscular) training loads in workouts for each player. (c) The importance of change of direction (COD) abilities and/or inter-effort recovery abilities on the V_{IFT} reached, since both directly influence the energetic cost of running during the test. As running economy determines the speed reached for a given $\dot{V}O_{2max}$ (and also ASR), a player with a poor COD will have a greater energetic cost of running and will present a lower $\dot{V}O_{2max}$ and, therefore, a slower V_{IFT} . Similarly, poor inter-effort recovery ability will be associated with a steeper $\dot{V}O_2$ /velocity relationship and/or premature muscular fatigue, resulting in slower running velocities. As for ASR, V_{IFT} takes into account COD and inter-effort recovery abilities and enables a more accurate tool to adjust training distances for these individual qualities.

Adapted from M. Buchheit, "The 30-15 Intermittent Fitness Test: 10 Year Review," *Myorobie Journal* 1 (2010): 1-19.



Figure 2.10 30-15 IFT performances across a selection of sports and playing standards for (a) men ($n=891$) and (b) women ($n=542$). Handball and soccer were chosen as the most complete samples, which were complemented with select group examples from different sports and standards. Contributions to the dataset provided by Cesar Meylan, Morgan Landrain, Daniel Peso, Alberto Pérez de Ciria, Andy Murray, Marcus Urban, Nick Poulos, Jack Nayler, Pat Sweeney, Shane Malone, Sebastian Knapić, Giacomo Schillaci, Antonio Dello Iacono, Simon Eustace, Heddy Arab, Thorsten Völler, Janeth Miranda, Francisco Javier Santos-Rosa Runao, Jamie Cook, Jaime Fernandez, Franck Kuhn, and David Hamilton.

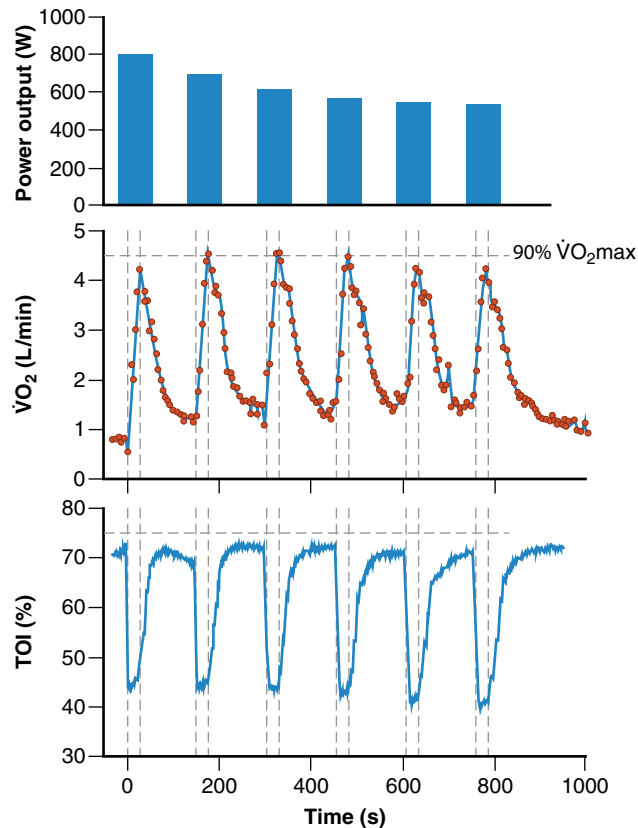


Figure 2.11 Power output, oxygen uptake ($\dot{V}O_2$), and muscle tissue oxygenation index (TOI) during a sprint interval training session (6×30 s all-out cycling sprints, 2 min passive recovery) completed by a well-trained cyclist. Note from this study that there was a progressive increase in muscle O_2 extraction with successive sprint repetitions, despite the decline in cycling power output, hence highlighting the increased reliance on aerobic metabolism as an interval training progresses in duration.

Data from M. Buchheit, C.R. Abbiss, J.J. Peiffer, and P.B. Laursen, "Performance and Physiological Responses During a Sprint Interval Training Session: Relationships With Muscle Oxygenation and Pulmonary Oxygen Uptake Kinetics," *European Journal of Applied Physiology* 112 no. 2 (2011): 767-779.

cessive sprint repetition (as shown by near-infrared spectroscopy), highlighting the progressive increase in aerobic energy system involvement that occurs as an all-out sprint interval training session progresses. This greater aerobic involvement drives fatigue-resistant adaptations that enhance subsequent all-out sprint performance.

Track-and-Field Approach

To program HIIT for endurance runners, coaches have traditionally used specific running speeds based on individual set times for distances ranging from 800 to 5000 m, but without using physiological markers such as heart rate, speeds associated with $\dot{V}O_{2max}$, or lactate turnpoints (7). Coaches and athletes have been, and still are, highly successful using this approach (see for example chapter 13). The attraction

of this method is that the entire locomotor profile (i.e., both maximal sprinting and aerobic speeds of the athlete) can be used to shape the HIIT session, so that each run can be performed in accordance with the athlete's (maximal) potential. While for short intervals (i.e., 10-60 s) the reference running time will be a percentage of the time measured over a maximal 100 to 400 m sprint, the speed maintained over 800-1500 m to 2000-3000 m can be used to calibrate longer intervals (e.g., 2-4 to 6-8 min). To illustrate this empirical programming approach, Pirie (97) provides session examples in athletes across different locomotor profiles for us. A sprinter capable of running 100 m inside 11 s might reasonably take 12 to 13 s for his 100 m training runs. A 400 m man under 50 s might cover 100 m in 14 to 15 s; the 800 m man under 1:53 and a 1500 m runner under 3:50 in 14 to 15 s; the longer-distance runner inside 14:30 or 10,000 m inside

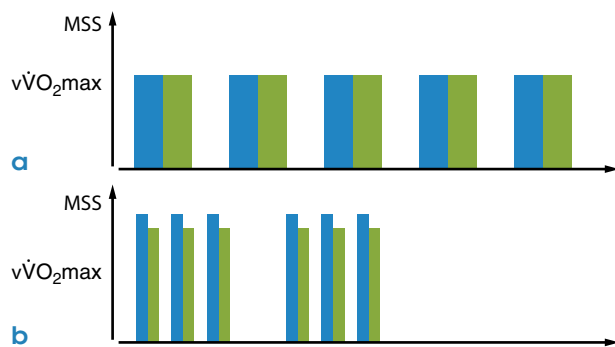


Figure 2.12 Running objectives during two track sessions for two athletes with different locomotor profiles. Runner A: V_{IncTest} 21.5 km/h, MSS 33.5 km/h; runner B: V_{IncTest} 21.5 km/h, MSS 29 km/h. (a) 12x400 m (only the first 5 repetitions are shown). As runs are performed around the V_{IncTest} , they perform the same work in 65 s. (b) 2x3x200 m. As runs are performed at a speed close to MSS, runner A (blue) performs these in 25 s, while runner B (green) performs these in 29 s.

31:00 in 16 to 17 s. For training at 200 m, the sprinter might run that distance in 25 to 26 s, the quarter miler in 28 to 30 s, 800 and 1500 m men in about 30 s, and long-distance runners in 33 to 34 s. The intervals between repetitions will, like those for the 100 m training, depend on the athlete's ability: if in the intermediate stage, 60 s; if a beginner, 75 s. After 3 or 4 mo, 40 repetitions should be reached (97). The disadvantage of this approach, however, is that it may be a bit harder for the coach to consciously manipulate the acute physiological load of the HIIT session and precisely target a specific adaptation (i.e., figure 2.12). Additionally, this approach tends to be reserved for highly experienced coaches and well-trained athletes, for whom target running times on several set distances are known. The translation and application of the track-and-field method to other sports is difficult. For example, how might a coach determine the expected 800 m run time for a 2.10 m tall basketball player who has never run more than 40 s continuously on a court before? Thus, using the track-and-field approach for nontrack-and-field athletes is unlikely to be appropriate, practical, or effective.

Game-Based HIIT or Small-Sided Games

Due to the technical and tactical requirements of team sports, and following the important principle of training specificity, game-based conditioning (i.e.,

so-called small-sided games, SSGs) (31, 37, 59, 73) or skill-based conditioning (60, 108) have received an exponential growth in interest (68). While understanding of the $\dot{V}O_2$ responses to game-based HIIT or SSG is limited (33, 37, 38), T (time) at $\dot{V}O_{2\max}$ during an SSG in national-level handball players was achieved for 70% of the session (i.e., 5:30 of the 8 min game) (33). Although the training effectiveness of such an approach has been shown (31, 49, 67, 73), SSGs have limitations that still support the use of less-specific (i.e., run-based) but more-controlled HIIT formats at certain times of the season or for specific player needs (figure 2.13). The acute physiological load of an SSG session can be manipulated by changing the technical rules (50), the number of players and pitch size (101), but the overall load cannot, by default, be precisely standardized. Within-player responses to SSGs are highly variable (poor reproducibility for blood lactate [coefficient of variation (CV): 15%-30%] and high-intensity running responses [CV: 30%-50%] (65, 66)), and the between-player variability in the (cardiovascular) responses is higher than more-specific run-based HIIT (68). During SSG in handball, average $\dot{V}O_2$ was shown to be inversely related to $\dot{V}O_{2\max}$ (33), suggesting a possible ceiling effect for $\dot{V}O_{2\max}$ development in fitter players. Additionally, reaching and maintaining an elevated cardiac filling pressure is believed to be necessary to improve maximal cardiac function (chapter 3) (48, 64). The repeated changes in movement patterns and the alternating work and rest periods during SSGs might therefore induce variations in muscular venous pump action, which can, in turn, limit the maintenance of a high stroke volume (SV) throughout the exercise and compromise long-term adaptations (71). Compared with generic run-based exercises, the $\dot{V}O_2$ to heart rate (HR) ratio, which can be used with caution as a surrogate of changes in SV during constant exercise when the arteriovenous O_2 difference is deemed constant (113), is also likely lower during SSGs (33, 37, 38). While this ratio is generally close to 1 during run-based long intervals (i.e., $\dot{V}O_2$ at 95% $\dot{V}O_{2\max}$ for HR at 95% of maximal $HR_{\max}$ (34, 102)), some have reported values at 79% $\dot{V}O_{2\max}$ and 92% $HR_{\max}$ during basketball drills (37) and at 52% $\dot{V}O_{2\max}$ and 72% $HR_{\max}$ during a 5-a-side indoor soccer game (38). This confirms the aforementioned possible limitations with respect to SV enlargement, and suggests that assessment of cardiopulmonary responses during SSGs (and competitive games) using HR may be misleading (87). Finally, the $\dot{V}O_2$

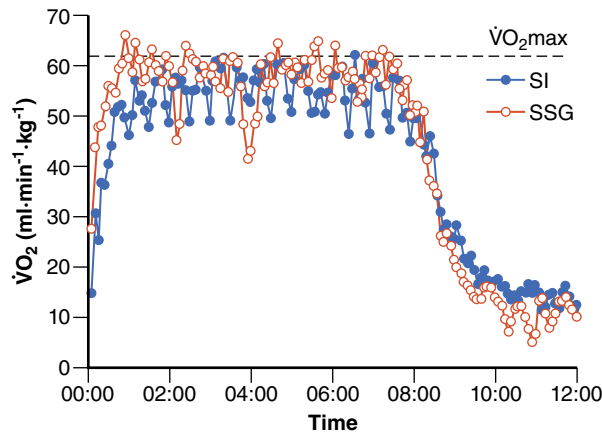


Figure 2.13 Example of the oxygen uptake ($\dot{V}O_2$) response obtained in the same player during an 8 min small-sided game (SSG) and a short-interval HIIT session (SI, 15 s at 95% of VIFT and 15 s passive). $\dot{V}O_{2max}$ refers to the maximal oxygen uptake measured during an incremental test (31).

vs. speed (33) and HR vs. speed (87) relationships also tend to be higher during SSGs compared with generic running, possibly due to higher muscle mass involvement. While this method of training is often considered to be highly specific, this is not always the case, since during competitive games players often have more space to run and reach higher running speeds (up to 85%-90% of maximal sprinting speed (36, 54, 88)) for likely similar metabolic demands. Finally, there is a very important aspect of such sport-specific HIIT that should not be overlooked. That is, in contrast with run-based exercises, the likely greater occurring neuromuscular load due to the frequently repeated accelerations, decelerations, and changes of direction is very difficult to predict and needs to be accounted for in terms of neuromuscular load programming (chapter 3). Game-based HIIT and SSG intricacies are expanded on in chapters 4 and 5.

As shown nicely by Hill-Haas et al. (68), the heart rate responses to various football training activities, including SSGs, are shown in figure 2.14. Notwithstanding the limitations of inferring metabolic demands with HR measurement, SSGs achieve high workloads (near 90% HR_{max}), only slightly less than circuit and HIIT efforts. Thus, SSGs are an effective tool in the practitioner's programming puzzle toolbox, so long as their advantages and limitations are realized. The efficacy of SSGs may be revealed by

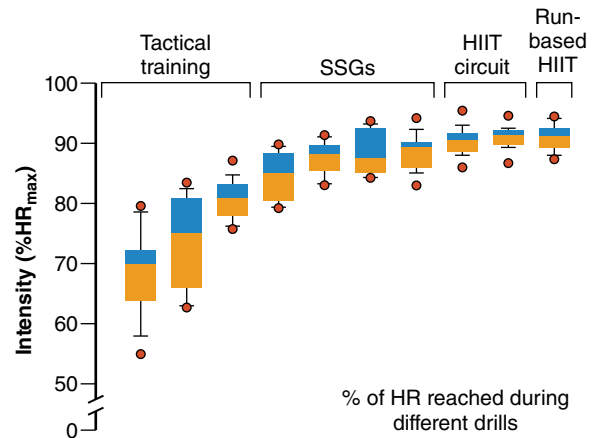


Figure 2.14 HR responses during tactical training, small-sided games (SSGs), circuit training and run-based HIIT.

Reprinted by permission of Springer Customer Service Centre GmbH from S.V. Hill-Haas, B. Dawson, F.M. Impellizzeri, and A.J. Coutts, "Physiology of Small-Sided Games Training in Football," *Sports Medicine* 41 no. 3 (2011): 199-220.

the fact that they tend to be used within nearly all team and racket sport practitioner application chapters in part II.

Summary

In summary, this chapter has highlighted the varying ways that HIIT is traditionally prescribed in practice by coaches to solve their performance puzzles. Using RPE-based prescription would be considered the universal HIIT practice, and is still used by most coaches today. Intervals calibrated with reference to the maximal aerobic speed and power are thought to be critical prescription components for many sports, and the upcoming chapter explains to us why that might be the case. Additionally, HR-based monitoring as opposed to prescription per se was shown to be the more efficacious method of use across HIIT. The somewhat forgotten anaerobic speed reserve, or upper capacity for high-intensity exercise above $v/p\dot{V}O_{2max}$, is an important factor to consider when calibrating supramaximal efforts, and the 30-15 intermittent fitness test was shown to be an effective means of measuring its capacity for appropriate HIIT prescription in team sports to target specific adaptations. Other methods of HIIT prescription that may be specific to the sport in general include the track-and-field approach and small-sided games for team sports.

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